Youcheng Li

- Peking University
- youchengli@stu.pku.edu.cn
- +86 13686820202
- youchengli.com

Born in June 2001 | CPC Member | Han Chinese

Profile

I am a Ph.D. student in Artificial Intelligence at the School of Intelligence Science and Technology , Peking University, advised by Prof. Liwei Wang . My research focuses on medical AI, especially applications of computer vision, generative models, and reasoning systems in biomedical imaging and drug discovery. I have published multiple papers in leading journals and conferences, including **Nature Biomedical Engineering**, **KDD**, and **MICCAI**, with several co-first-author contributions. On the industry side, I co-founded **Beijing Yisuo Technology Co., Ltd.** and serve as CTO, leading the development of the **AI Scientist** agent system, which has been deployed at leading Grade-A tertiary hospitals, top universities, and publicly listed pharmaceutical companies in China and abroad. I also helped the company complete tens of millions of RMB in early-stage financing. Mission: build medical AI systems that truly reach practice and serve human health and life science research.

Education

Xi'an Jiaotong University — *Sep 2019 – Jul 2023*
B.Eng. in Artificial Intelligence

- GPA: 4.0/4.3
- **Scholarships:** National Scholarship (top 1%), Mitacs Globalink Research Award (top 1%), China Scholarship Council Scholarship (top 1%), MEGVII Scholarship (top 3%), Zheng Guobin Scholarship (top 3%)

Peking University — *Sep 2023 – Present*
Ph.D. in Engineering, Artificial Intelligence — *Expected Jul 2028*

- Advised by Prof. Liwei Wang; research focus on medical AI, generative models, and AI agent systems for biomedical applications
- **Scholarship:** National Scholarship (top 1%)

Experience

Co-founder & CTO — *Beijing*
Beijing Yisuo Technology Co., Ltd. — *Jun 2025 – Present*

- As co-founder and CTO, lead the company's overall technical strategy and product roadmap. The core product is the **AI Scientist** software-hardware agent system for biomedical R&D and scientific research.
- Built and led cross-functional engineering teams across front-end, back-end, and AI agent development, driving timely delivery of the MVP and subsequent versions.
- Led end-to-end R&D of the agent system, including architecture design, prompt engineering and optimization, and toolchain integration, significantly improving system performance and stability.
- Led the Physical AI technology roadmap, system design, and proof-of-concept demo development.
- Designed the technical roadmap for scientific world models, including model training and evaluation.
- Drove commercial deployment of the AI Scientist platform at leading Grade-A tertiary hospitals, top universities, and publicly listed pharmaceutical companies in China and abroad.
- Deeply involved in business development and financing, helping the company complete tens of millions of RMB in seed financing and supporting subsequent fundraising efforts.

Algorithm Research Intern — *Beijing*
Yizhun Intelligent Technology (Beijing) Co., Ltd. — *Jan 2023 – Jun 2025*

- Led deep learning algorithm R&D for breast cancer screening and diagnosis; related work was published at **MICCAI 2023** and in **Nature Biomedical Engineering** as co-first-author contributions.
- Developed a real-time dynamic lesion detection algorithm for breast ultrasound, achieving **sensitivity >95%** and reducing inference latency by **15%** through model compression and hardware acceleration.
- Designed a deep-learning-based intelligent diagnosis algorithm for ductal carcinoma in situ (DCIS), winning the **National First Prize** in the 2nd National Digital Health Innovation Application Competition hosted

by the National Health Commission.

- Built a foundational generative model for breast ultrasound, enabling high-fidelity synthetic data generation and supporting large-scale training for multiple downstream diagnostic tasks.

Research Assistant — *Canada (Remote)*
Western University — *Sep 2022 – Jul 2023*

- Supported by Mitacs and the China Scholarship Council, conducted research on cell segmentation for spatial transcriptomics under the supervision of Prof. Pingzhao Hu; proposed a multi-scale manifold learning method and published the work in **PLOS Computational Biology** as first author.
- Implemented the method in Python/PyTorch and conducted systematic benchmarking against mainstream methods including Cellpose and StarDist.

Embedded Engineering Assistant Intern — *Shenzhen*
Shenzhen Anke High-tech Co., Ltd. — *Jul 2020 – Sep 2020*

- Participated in medical device hardware development, responsible for circuit debugging, firmware optimization, and embedded system integration testing on ARM platforms.

Publications

Note: † indicates co-first authorship.

Dai D†, Liu B†, **Li Y**†, Yu H†, et al. MammoExpert: Benchmarking Chain-of-Thought Reasoning in Mammography Diagnosis[C]//**KDD 2026 AI4Sciences Track**. 2026. *(Accepted, co-first author)*

Yu H†, **Li Y**†, Zhang N, et al. A Foundational Generative Model for Breast Ultrasound Image Analysis[J]. **Nature Biomedical Engineering**, 2025. *(Co-first author)*

Yu H†, **Li Y**†, et al. A Chain-of-thought Reasoning Breast Ultrasound Dataset Covering All Histopathology Categories[J]. **Scientific Data**, 2025. *(Co-first author)*

Li Y, Lac L, Liu Q, et al. ST-CellSeg: Cell segmentation for imaging-based spatial transcriptomics using multi-scale manifold learning[J]. **PLOS Computational Biology**, 2024, 20(6): e1012254. *(First author)*

Yu H, **Li Y**, Wu Q L, et al. Mining negative temporal contexts for false positive suppression in real-time ultrasound lesion detection[C]//**MICCAI 2023**. Cham: Springer Nature Switzerland, 2023: 3–13.

Yu H, **Li Y**, Zhang N, et al. Knowledge-driven AI-generated data for accurate and interpretable breast ultrasound diagnoses[J]. arXiv preprint arXiv:2407.16634, 2024.

Yan H, Dai C, Xu X, et al. Using artificial intelligence system for assisting the classification of breast ultrasound glandular tissue components in dense breast tissue[J]. **Scientific Reports**, 2025, 15(1): 11754.

Honors and Awards

- **National First Prize**, 2nd National Digital Health Innovation Application Competition, hosted by the National Health Commission, 2024
- **National Scholarship** (top 1%), Peking University, 2024
- **National Scholarship** (top 1%), Xi'an Jiaotong University, 2022
- **Mitacs Globalink Research Award** (top 1%), Canada, 2022
- **China Scholarship Council Scholarship** (top 1%), 2022

Skills

Programming Languages: Python, C++, JavaScript, SQL

Deep Learning and Models: PyTorch, model compression and quantization, TensorRT, ONNX, generative models (diffusion models/GANs)

AI Agents and Large Language Models: LangChain/LangGraph, prompt engineering, multi-agent system design, RAG

Engineering: Docker, Git, Linux, TensorRT, ONNX, embedded systems (ARM/RTOS)

Research Areas: medical image analysis, foundational generative models, biomedical AI, spatial transcriptomics